

MANIKANTA GOPI

Senior Backend & Platform Engineer

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PROFESSIONAL SUMMARY

Backend and platform engineer with 4+ years of production experience building and operating backend systems, distributed workloads, cloud infrastructure and Kubernetes platforms. Focused on infrastructure cost optimization, reliability, performance, cloud migration and asynchronous systems across AWS, GCP and Azure.

ENGINEERING IMPACT

- \$18k to \$1.8k/month - reduced video-processing infrastructure by moving Lambda + FFmpeg workloads to in-house Kubernetes workers.
- \$500/month - reduced database costs by identifying resource-intensive queries and changing the surrounding systems.
- \$400/month - reduced Kubernetes compute costs using Karpenter with Spot + On-Demand capacity.
- \$800 saved in May - added S3 expiry for ephemeral media and prevented uncontrolled storage growth.
- Zero-downtime migration - led Azure AKS to AWS EKS migration using Terraform and service-by-service cutover.
- Zero-downtime queue migration - moved queueing infrastructure from Redis to Dragonfly.

PROFESSIONAL EXPERIENCE

Senior Software Engineer | Circleapp Online Services (Praja)

Apr 2025 - Present

- Own GitOps workflows for ~24 production services on AWS EKS, including Helm, environment configuration and automated image updates.
- Reduced video-processing infrastructure from ~\$18k/month to ~\$1.8k/month and improved workload economics with Kubernetes workers.
- Optimized database and S3 lifecycle usage; introduced Karpenter Spot + On-Demand capacity and migrated Redis queueing to Dragonfly without downtime.
- Moved background removal from a hosted vision API to a self-hosted ONNX GPU service in Kubernetes.

Software Engineer | Circleapp Online Services (Praja)

May 2022 - Mar 2025

- Built and operated real-time messaging with NestJS, Socket.IO, Redis pub/sub, MongoDB and BullMQ.
- Led Azure AKS to AWS EKS production migration using Terraform, with service-by-service cutover and no maintenance window.
- Implemented subscription lifecycle workflows including charge scheduling, grace periods, refunds and idempotent payment handling.
- Built asynchronous poster and video generation workflows supporting the paid tier; published sidekiq-assured-jobs.

Software Engineer Intern | Circleapp Online Services (Praja)

Dec 2021 - Apr 2022

- Shipped Flutter chat features and built the Rails API endpoints supporting them.

Machine Learning Intern | Continual Engine

2021

- Built image-to-text models in PyTorch, including autoencoders for feature extraction, training and evaluation.

SELECTED PROJECTS

Seeker - Search Engine from First Principles:

Built an inverted index, BM25, Levenshtein automata, FSTs, BKD trees and a byte-level segment format in ~20,000 lines of TypeScript; documented the work in a 43-chapter book.

Media Processing Platform:

Built KEDA-scaled media workers driven by queue depth for poster and video generation using NestJS, FFmpeg, Puppeteer, Redis and S3.

Real-time Messaging:

Designed cross-instance fan-out with Socket.IO and Redis pub/sub, MongoDB persistence and BullMQ for deferred work.

Self-hosted Vision Service:

Deployed ONNX Runtime GPU inference in Kubernetes and validated a face-detector replacement against 200 real production images before shipping.

TECHNICAL SKILLS

Languages: Ruby, TypeScript, Go, Python, Dart, Bash, SQL

Backend: Ruby on Rails, NestJS, Sidekiq, BullMQ, Socket.IO, Async, Gin, GORM, REST API design

Distributed Systems: asynchronous processing, message queues, pub/sub, fan-out, idempotency, background jobs

Infrastructure: Kubernetes, AWS EKS, Helm, Terraform, Docker, KEDA, Karpenter, Flux, ArgoCD, APISIX

Data: MySQL, PostgreSQL, Redis, MongoDB, OpenSearch, S3

Cloud & Observability: AWS, GCP, Azure, CircleCI, GitHub Actions, Prometheus, Grafana, Loki, New Relic

EDUCATION & OPEN SOURCE

B.Tech, Computer Science | Rajiv Gandhi University of Knowledge Technologies | 2022 | GPA 9.3/10

sidekiq-assured-jobs - open-source Ruby gem for recovering in-flight Sidekiq jobs after worker termination.